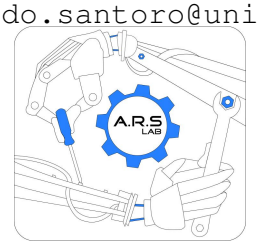
Fuzzy Logic

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Advanced Robotics and Autonomous Systems

Notes

Fuzzy Logic

Fuzzy Logic is a type of logic that does not consider only **false** and **true** (strong truth) to concepts but associates (to concepts) a **degree of truth** represented as a real number $\in [0, 1]$

Notes

Standard Logic vs. Fuzzy Logic

Standard Logic

The cup is hot

In standard logic, this sentence may be either false or true

Fuzzy Logic

The cup is hot

In fuzzy logic, this sentence has a degree of truth $\in [0, 1]$, thus making able to map something like very hot, hot, not so hot, etc.

Notes

Standard Logic vs. Fuzzy Logic

Fuzzy Sets

The degree of truth is defined through **fuzzy sets**.

A fuzzy set of a concept C is a function $\mu_C(x) \in [0, 1]$ that gives the degree of truth for concept C of a variable x of our universe.

Example

The cup is hot

- The concept is *hot*
- The variable is *CupTemperature*
- The fuzzy set is the map $\mu_{hot}(CupTemperature)$

Function $\mu_C(x)$ is also called **membership function**

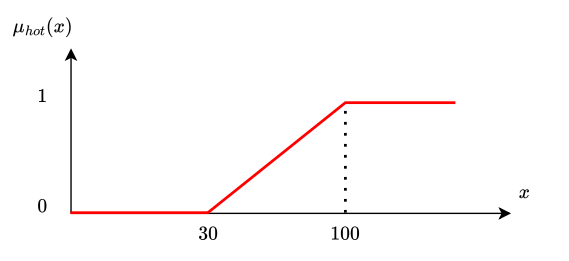
Notes

Fuzzy Set

Fuzzy sets are in general expressed as **linear functions**

Example of a Fuzzy Set

degree of truth of the concept *hot*:



$$\begin{aligned} \mu_{hot}(x) &= 0 & x < 30 \\ \mu_{hot}(x) &= \frac{(x-30)}{70} & 30 \leq x \leq 100 \\ \mu_{hot}(x) &= 1 & x > 100 \end{aligned}$$

Notes

Fuzzification

Variables, Memberships, Crisp and Fuzzy Values

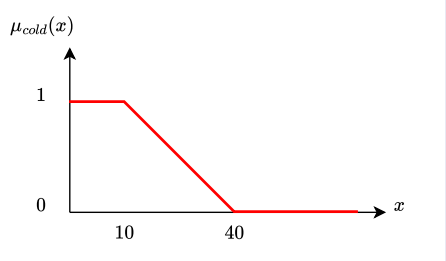
- A **variable** is something that represents a certain quantity in our universe, e.g. *CupTemperature*
- When the variable is considered in its universe, it is called **crisp** and its value is the **crisp value**, e.g. $CupTemperature = 37$
- The **membership function** $\mu_C(x)$ defines a concept C to be applyed to crisp variable x
- The application of the membership function to a crisp variable is called **fuzzification** and results in **fuzzy value**, i.e. $\mu_{hot}(CupTemperature) = 0.1$

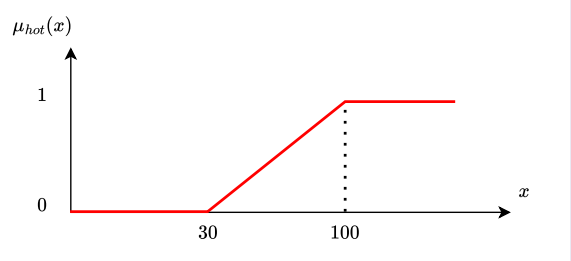
Notes

Fuzzification

Variables, Memberships, Crisp and Fuzzy Values

- A same (crisp) variable can be associated to different concepts, each with its relevant fuzzy value, because e.g. a temperature can be *not so hot* and also *not so cold*





CupTemperature	μ_{cold}	μ_{hot}
10	1	0
20	0.5	0
30	0	0
35	0.17	0.71
45	0	0.22

Notes

Fuzzy Reasoning and Rules

Notes

Fuzzy Rules

Fuzzy Rules

A fuzzy rule is a declaration of the form:

If *sentence1* Then *sentence_out*

If *sentence1* AND *sentence2* Then *sentence_out*

If *sentence1* OR *sentence2* Then *sentence_out*

The **degree of truth** of *sentence_out* depends on the degrees of truth of antecedents and the operator used (AND, OR)

Notes

Fuzzy Rules

Example

If temperature is hot Then fan speed is fast

We have two crisp variables:

- temperature, input variable
- fan speed, output variable

and two concepts, or fuzzy sets:

- hot
- fast

Notes

Fuzzy Rules

Example Algoritm

If temperature is hot Then fan speed is fast

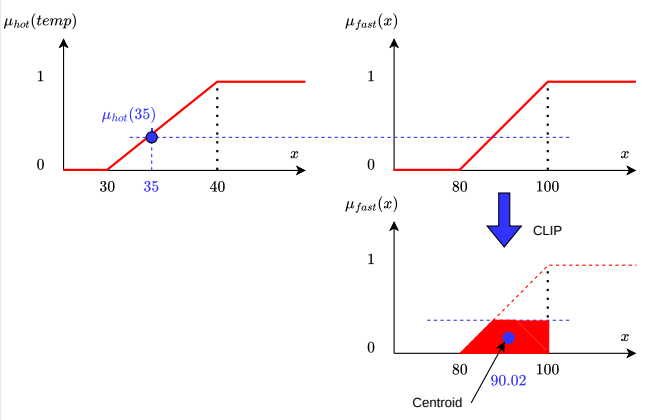
1. Define the fuzzy sets hot and fast
2. Assign the input value to temperature
3. **Fuzzify** the temperature
4. Apply the resulting fuzzy value to fast
5. **Defuzzify** concept fast thus resulting in a (crisp) value of variable fan speed, i.e. $\mu_{fast}^{-1}(fanspeed)$

Notes

Rule Computing and Defuzzification

Clipping

If temperature is hot Then fan speed is fast



1. Compute $\mu_{hot}(temperature)$
2. Clip the fast set at $\mu_{hot}(temperature)$
3. Defuzzify by extracting the abscissa of the **centroid**

Notes

Fuzzy Rules

Many Rules

A single rule is not so usefull, better having different rules that act on different concepts:

$R1$:**If temperature is hot Then fan speed is fast**
$R2$:**If temperature is warm Then fan speed is medium**
$R3$:**If temperature is cold Then fan speed is slow**

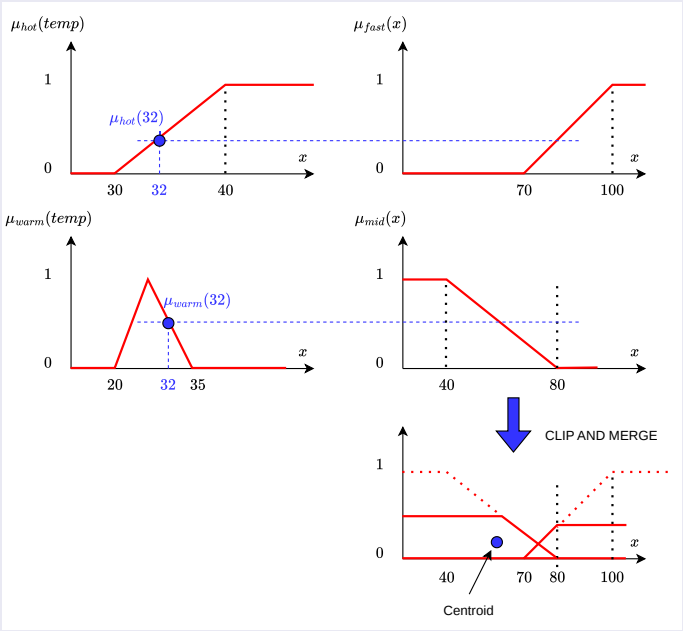
1. Gather the fuzzy sets associate to the same variable, properly clipped or scaled, i.e. $\mu_{fast}(fanspeed), \mu_{medium}(fanspeed), \mu_{slow}(fanspeed)$
2. Merge all of these sets (set union)
3. Compute the centroid of the resulting set
4. Extract the abscissa

Notes

Fuzzy Rules

Many Rules

*R*1 :**If temperature is hot Then fan speed is fast**
*R*2 :**If temperature is warm Then fan speed is medium**



Notes

Fuzzy Rules

Connectives

$R1$: **If temperature is hot AND humidity is high Then fan speed is fast**
$R2$: **If temperature is warm OR humidity is low Then fan speed is medium**

AND Connective

The fuzzy value to apply to the conseguent is the minimum of the values of antecedents, e.g.:

$$R1 : min\{\mu_{hot}(temp), \mu_{high}(hum)\}$$

OR Connective

The fuzzy value to apply to the conseguent is the maximum of the values of antecedents, e.g.:

$$R2 : max\{\mu_{warm}(temp), \mu_{low}(hum)\}$$

Notes

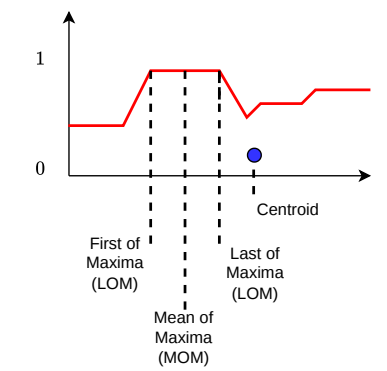
Defuzzification

Defuzzification Methods

- **center of gravity** (centroid):

$$x^* = \frac{\sum x_i \mu_i(x_i)}{\sum \mu_i(x_i)}$$

- **first of maxima** *FOM*
- **last of maxima** *LOM*
- **mean of maxima** $MOM = \frac{FOM+LOM}{2}$



Notes

Fuzzy Rules: Mamdani vs. Sugeno Approach

Mamdani

The conseguent is a fuzzy set:

$R1$: **If temperature is hot AND humidity is high Then fan speed is fast**
$R2$: **If temperature is warm OR humidity is low Then fan speed is medium**

Sugeno

The conseguent is a function:

$R1$: **If temperature is hot AND humidity is high Then fan speed is** $(temperature - 25) * 0.1$
$R2$: **If temperature is warm OR humidity is low Then fan speed is** $(temperature - 25) * 0.1 + humidity/10.0$

Output is the weighted mean where the weight is the fuzzy value of the rule:

$$out = \frac{\sum out_i \mu_i}{\sum \mu_i}$$

Notes

Fuzzy Control

Notes

Fuzzy Control

Fuzzy Control

- A **fuzzy controller** is a controller that uses fuzzy rules to determine the output value of the driving signal
- It can use concepts such as error is big/small/etc.
- It can combine several input variables, e.g.:
  error is big AND speed is low negative
- It is a sort of PID with variable gains

Notes

Example: A Fuzzy Speed Controller

The Classical PI Approach

By using PIDs we need:

- A proportional part when the error is big, in order to react immediatelly
- A integral part that accumulates the error and generate a proper value when the error is small

A possible fuzzy approach

- Use only an integrator
- **integrate fast** when the error is big
- **integrate slow** when the error is low

Notes

Example: A Fuzzy Speed Controller

A possible fuzzy approach

Let's use a variable accelerate as output

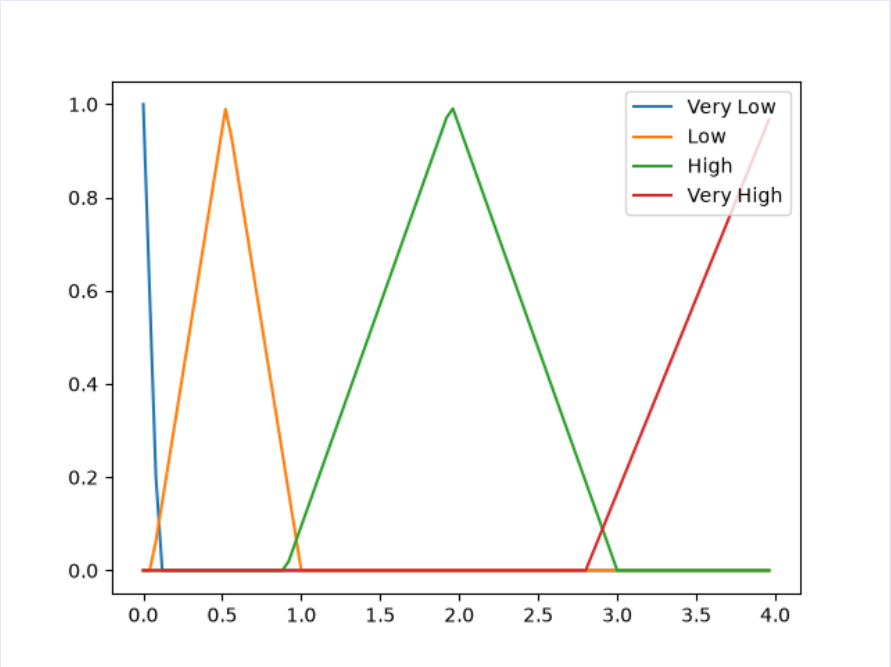
*R*1 : **If error is very low Then accelerate very slow**
*R*2 : **If error is low Then accelerate slow**
*R*3 : **If error is high Then accelerate fast**
*R*4 : **If error is very high Then accelerate very fast**

Rules are simple, the key is in the definition of fuzzy sets.

Notes

Example: A Fuzzy Speed Controller

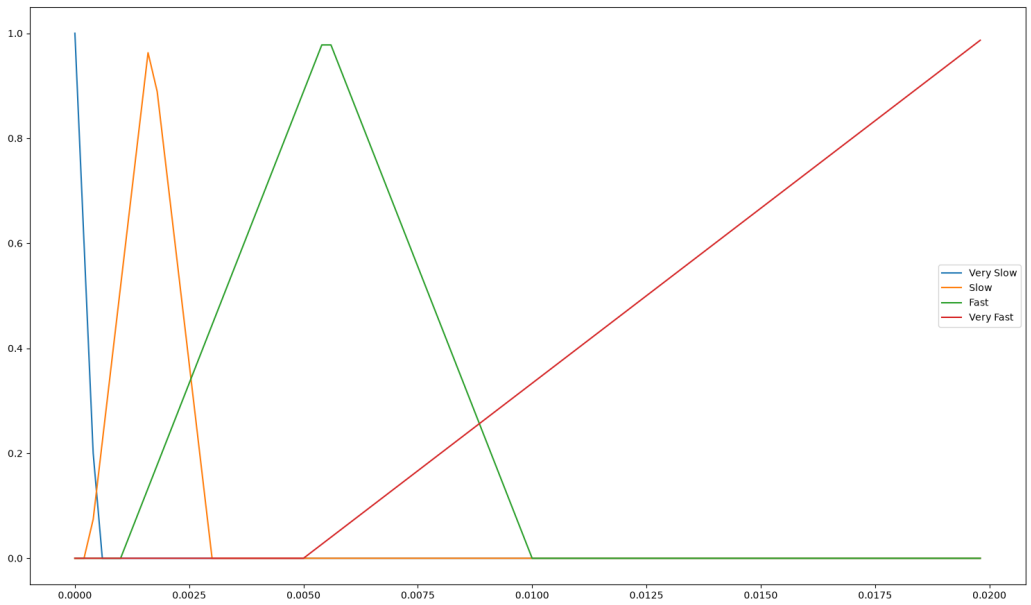
Fuzzy Sets: Error



Notes

Example: A Fuzzy Speed Controller

Fuzzy Sets: Acceleration



Notes

Fuzzy Position Control

Notes

Fuzzy Position Control

A possible fuzzy approach

Let's use two input variables:

- $Error = TargetPosition - CurrentPosition$
- $\Delta error = Error - PrevError$

one output variable:

- $Push$

And for each variable, let's introduce five fuzzy sets/concepts:

- **Negative Big (*NB*)**
- **Negative Small (*NS*)**
- **Zero (*ZE*)**
- **Positive Small (*PS*)**
- **Positive Big (*PB*)**

Notes

Fuzzy Position Control

The Control Matrix

		$\Delta Error$				
		NB	NS	ZE	PS	PB
Error	NB	NB	NB	NB	NB	NS
	NS	NB	NS	NS	ZE	ZE
	ZE	NS	NS	ZE	PS	PS
	PS	ZE	ZE†	PS	PS	PB‡
	PB	PS*	PB	PB	PB	PB

- $Error = TargetPosition - CurrentPosition$
- $\Delta error = Error - PrevError$

Some Rules

*: if *Error* is pos big and $\Delta Error$ is neg big then push is pos small
†: if *Error* is pos small and $\Delta Error$ is neg small then push is zero
‡: if *Error* is pos small and $\Delta Error$ is pos big then push is pos big

Notes

Fuzzy Control vs. PID Control

PID Control

- Analytical approach
- Low computation complexity
- Tuning requires system model, if automatic methods are used
- Fixed Gains
- Suitable for linear systems

Fuzzy Control

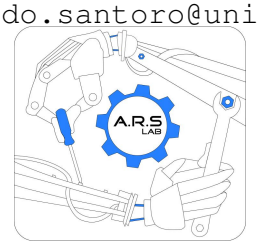
- Experience-based approach
- High complexity
- Tuning does not require system model (empirical)
- Flexible (varible) gains
- Suitable for non linear systems

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